

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459924.686 +/- 0.027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.086 +/- 0.032
Transit depth (Rp/Rs)^2	0.73 +/- 0.55 [%]
Orbital Inclination (inc)	89.29 +/- 2.41
Airmass coefficient 1 (a1)	1.027 +/- 0.016
Airmass coefficient 2 (a2)	-0.027 +/- 0.088
Scatter in the residuals of the lightcurve fit is	1.53 %
Best Comparison Star	None
Optimal Aperture	6.15
Optimal Annulus	16.93
Transit Duration (day)	0.111 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.086 ± 0.032 vs 0.1066 (deviation 0.64σ)
Duration fit vs published	0.111 d vs 0.1196 d (deviation 0.66σ)
Mid-time O–C	17.7 min
Depth SNR	2.7
Residual scatter	1.53 %

Light curve

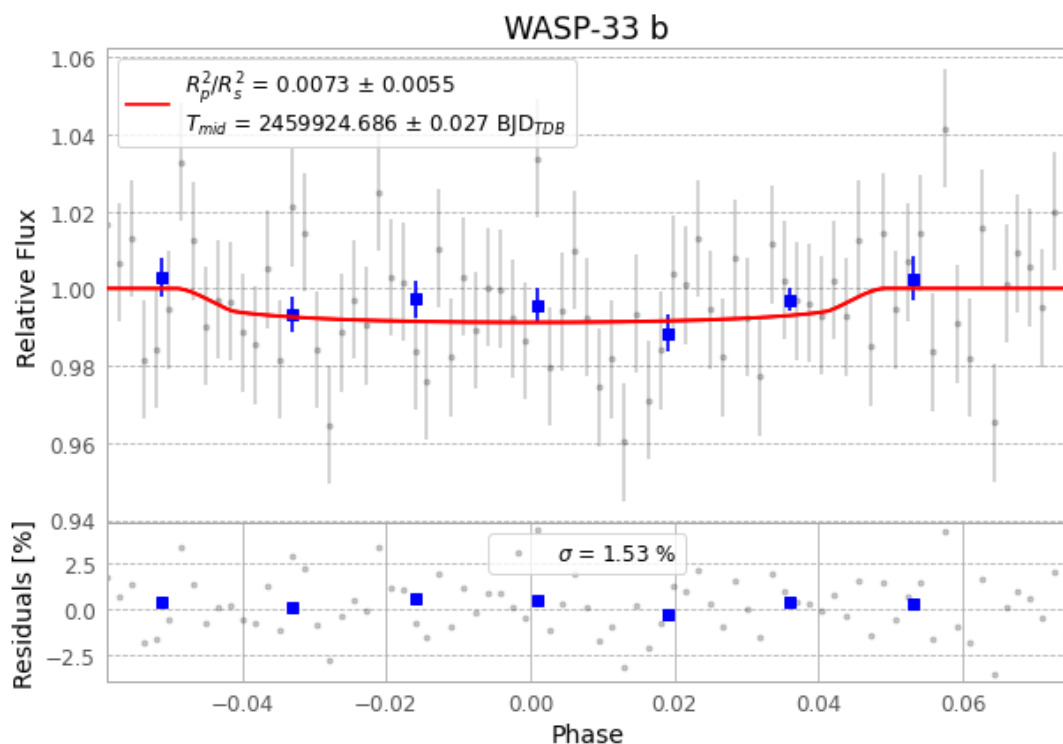


Figure 1 — FinalLightCurve_WASP-33 b_2022-12-10.png (SHA-256 560b825eab4bca16...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [446, 150], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7c33bfb01c009635...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8d12984

Data provenance

79 FITS frames (52 MB), WASP-33221211023615.FITS → WASP-33221211063016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-33-221211.log (2f17dd465cfa...), FinalLightCurve_WASP-33 b_2022-12-10.png (560b825eab4b...), FinalLightCurve_WASP-33 b_2022-12-10.csv (b10dad3e469f...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 14:11Z.